IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE PART IV: MEng and ACGI

POWER SYSTEM DYNAMICS STABILITY AND CONTROL

Friday, 17 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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The Questions

1.

a)

i) Explain the difference between lightning and switching surges in power systems.

[4]

ii) How do these phenomena impact the operational stability of a power system ?

[4]

iii) What are stator transients in the context of power system stability ?

[2]

iv) How are these stator transients modelled for electromechanical stability studies involving synchronous machines ?

[2]

v) How are these stator transients modelled for stability studies involving asynchronous machines used for wind power applications ?

[2]

b)

i) What is electrical resonance in the context of power system operation ?

[4]

ii) Where are such resonance phenomena observed ?

[2]

ii) How sub-synchronous torsional interaction (SSTI) is different from electrical resonance ?

[3]

iii) How has more converter interfaced generation (CIG) in electric power systems influenced the degree of presence and impacts of both the above phenomena ?

[2]

[25 marks]

2.

a)

i) Some synchronous generators are of 2-pole while some are 4, 6 , 8 or even higher poles – Explain the reasons with reference to suitable applications. [3]

ii) Usually the stator of a synchronous generator is laminated whereas the rotor is solid. However, in many cases additional windings are housed in the rotors. What are the purposes of these windings and how do they impact the stability performance of the generator as well as that of the power system? [2+2]

iii) Synchronous machines with larger air gap have better synchronisation capability. But in practice, a very small or minimum air-gap construction is adopted. Explain the reason behind this, and also narrate how synchronisation capability is ascertained in low air-gap machines. [2+2]

b) A round rotor star connected synchronous generator is connected to the grid at 20 kV (line to line) voltage. The rated power of 500 MW, power factor is 0.85 (lagging). The synchronous reactance is 1.50 p.u. on its rated base of 588 MVA. The resistance is negligible. Find out:

i) the rated line current in kA [2]

ii) the excitation (phase) voltage in kV [3]

iii) load angle (δ) in degree [2]

iv) steady state stability margin in degree [1]

iv) total reactive power in MVAR [3]

c) The excitation voltage is decreased by 20%

i) Find out the change in stability margin in degree with respect to the rated excitation voltage. [3]

[25 marks]

3.

a)

i) Why does one need to move to a rotational reference frame for synchronous machine dynamic modelling ?

[2]

ii) How does a rotational reference frame help in voltage, current and power calculation in synchronous machine dynamic analysis ?

[3]

iii) Why, in a multi-machine power system context, the variables in rotational reference frame need further transformation ?

[3]

iv) In lagging power factor operation, the transformed current and voltage in d-axis oppose the field current and exciation voltage. Explain the reasons.

[4]

v) What impact will such a lagging power factor operation have on the generator small signal as well as large signal (transient) stability ?

[2+2]

b)

i) What are the basic functional and constructional differences between the Type 3 (DFIG) and Type 4 (full converter) wind turbine generators (WTGs) ?

[4]

ii) Which of the active stall and active pitch control in modern WTGs is the preferred choice and why ?

[1+2]

iii) What is the importance of optimal tip speed ratio (TSR) in the context of WTG operation ?

[2]

[25 marks]

4.

a)

i) Define co-efficient Cp in wind power conversion.

[2]

ii) What design and operating parameters influence Cp ?

[2]

iii) Starting from the first principle of aerodynamics, and including the assumptions, establish that the Betz limit is 16/27.

[7]

b)

i) What is the fundamental difference between pitch and stall control ?

[2]

ii) How can WTG offer frequency response service at subrated wind speed?

[3]

iii) How is it possible to have net power production from Type 3 WTG below the synchronous speed ?

[3]

iv) For Type 4 WTG, why is a gearbox not mandatory ?

[2]

v) Why are turbine blades not controlled to track fast and random wind speed variation ?

[2]

vi) Why in a wind farm, cables are never loaded beyond less than half of their rated capacity even at the rated wind speed ?

[2]

[25 marks]